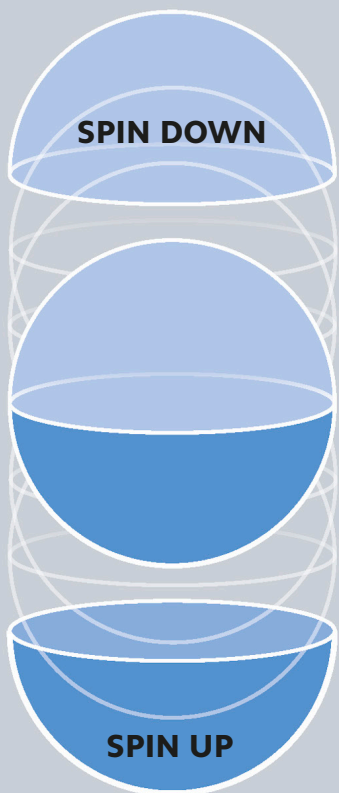
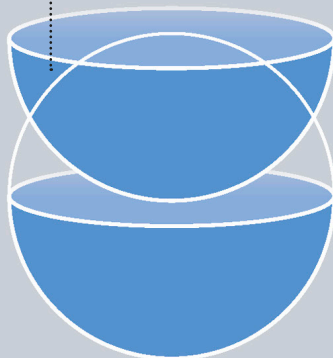




**REJECTED
PAIRING**

Particles with identical spin cannot share the same state—one must always go somewhere else.



Particle pairs

Two otherwise identical particles can have the same energy, momentum, and position if their spins are complementary.

NO ROOM FOR TWO

One of the most important rules underlying the structure of matter, the Pauli exclusion principle prevents fermions from falling into completely identical quantum states. This generates a form of pressure that keeps particles apart even when other repulsive forces such as electromagnetism fail (for instance, inside superdense collapsed stars known as white dwarfs and neutron stars). It also explains why only two electrons (with up and down spins) can occupy the same orbital subshell within an atom (see p.31).